



# COMPENDIUM OF INDIAN STANDARDS ON **FIRE EXTINGUISHERS**

Prepared By:

**CIVIL  
ENGINEERING  
DEPARTMENT**



**BUREAU OF INDIAN STANDARDS  
NEW DELHI**

# INTRODUCTION

Fire safety is a fundamental aspect of protecting lives, property, and critical infrastructure across all sectors of society. With increasing urbanization, industrial growth, and the complexity of built environments, the demand for reliable and effective fire protection measures has never been more urgent. Portable and wheeled fire extinguishers serve as the first line of defence against incipient fires, making robust standards for their design, performance, installation, and maintenance essential for public safety and regulatory compliance. The Bureau of Indian Standards (BIS), through its dedicated technical committees, has developed a comprehensive suite of Indian Standards to address every facet of fire extinguisher technology from material specifications and manufacturing practices to rigorous testing protocols and guidelines for selection, placement, and upkeep.

This compendium is designed as a ready reference for a wide range of stakeholders, including manufacturers, safety engineers, facility managers, regulators, policy makers, researchers, and academic institutions. It consolidates the latest Indian Standards covering all major types of fire extinguishers such as water, foam, powder, carbon dioxide, clean agents, Halon, and specialized extinguishers for Class D and aviation use along with guidelines for their installation and ongoing maintenance as per IS 2190.

By bringing together these essential standards in a single volume, the compendium aims to foster greater awareness, facilitate compliance, and promote uniformity in fire safety practices across the country. It supports national priorities in disaster resilience, workplace safety, and sustainable development by ensuring that fire extinguishers in India meet the highest benchmarks of quality, reliability, and performance.

The publication of this compendium underscores BIS's enduring commitment to advancing fire safety and providing accessible, high-quality standardization resources to the nation's building, industrial, and public safety sectors.

## **IS 15683: 2018 PORTABLE FIRE EXTINGUISHERS — PERFORMANCE AND CONSTRUCTION — SPECIFICATION (FIRST REVISION)**

This standard specifies the requirements for the performance, reliability, and safety of portable fire extinguishers across various types, including water-based, foam-based, powder (ABC/BC/D type), carbon dioxide, clean agents, and water mist extinguishers. It mandates that extinguishing media conform to relevant IS standards, with strict guidelines for raw materials, fill densities, and propellants to ensure both safety and effectiveness. The construction of extinguisher bodies must use durable, corrosion-resistant materials like steel, aluminium, or approved composites, and all units must undergo rigorous pressure, stress, and safety testing. Each extinguisher is subject to comprehensive type and routine tests such as pressure cycling, burst, leakage, vibration, and fire performance tests before certification. These provisions guarantee that only extinguishers meeting all criteria for durability, safety, and fire-fighting efficacy are approved for use.

Portable fire extinguishers are essential safety devices for immediate response to incipient fires in homes, offices, factories, vehicles, and public spaces. They are selected and maintained based on specific fire hazards present, providing targeted protection for solids, liquids, gases, metals, or cooking oil fires.

## **IS 16018: 2012 WHEELED FIRE EXTINGUISHERS — PERFORMANCE AND CONSTRUCTION — SPECIFICATION**

This standard outlines the essential requirements for the safety, reliability, and performance of wheeled fire extinguishers, applicable to models with a fully charged mass between 17 kg and 450 kg. It covers extinguishers using water, dry powder, foam, carbon dioxide, or clean agents, with strict specifications for raw materials, fill densities, and propellants to ensure effectiveness and safety. The construction mandates robust, corrosion-resistant bodies made from carbon steel, stainless steel, or aluminium, and all critical components are designed for pressure safety and durability. Each extinguisher must pass rigorous tests for discharge range, temperature and corrosion resistance, pressure integrity, fire performance, and operational durability, including simulated rough handling. Clear markings and operating instructions are required for safe and correct use by trained personnel.

**Applicability and Use:** Wheeled fire extinguishers are ideal for industrial sites, warehouses, refineries, and large outdoor storage areas where significant fire risks exist and extended discharge capacity is needed. They provide powerful, mobile fire suppression for Class A, B, C, and D fires, enabling rapid and effective response by trained users.

## IS 11108: 1984 SPECIFICATION FOR PORTABLE FIRE EXTINGUISHERS — HALON 1211 TYPE (Reaffirmed in December 2020)

This standard specifies the requirements for portable fire extinguishers of the Halon 1211 type, detailing the materials, construction, anti-corrosive treatments, and performance tests necessary to ensure safety and reliability. The extinguisher body and components are made from corrosion-resistant materials such as carbon steel, bronze, brass, stainless steel, and rubber, with Halon 1211 as the extinguishing agent charged with dry nitrogen. Designs may be upright or inverted, featuring secure valve mechanisms and robust construction for safe and effective operation. Each extinguisher undergoes rigorous testing including hydrostatic, leakage, drop, burst pressure, and performance tests to verify discharge efficiency, jet throw, and corrosion resistance. These extinguishers are compact, non-conductive, and deliver a clean agent, making them especially suitable for sensitive and confined environments.

**Applicability and Use:** Halon 1211 portable fire extinguishers are ideal for use in control rooms, electronic equipment areas, and confined spaces where electrical or flammable liquid fires (Class B and C) may occur. Their clean, non-conductive discharge makes them safe for sensitive environments and critical assets.

## IS 11833: 1986 SPECIFICATION FOR DRY POWDER FIRE EXTINGUISHER FOR METAL FIRES (Reaffirmed in September 2022)

This standard defines the requirements for dry powder fire extinguishers specifically designed for Class D fires involving burning metals, available in capacities ranging from 10 to 100 kg. The extinguisher bodies are constructed from mild steel, with components such as the cap, nozzle, and fittings made from corrosion-resistant materials like leaded tin bronze, brass, aluminium alloy, or rubber, and filled with a specialized dry powder agent. The design includes features like a syphon tube, drain plug, applicator, and towing handle, with larger units mounted on wheeled carriages for mobility. Low-velocity discharge through a hose and nozzle prevents scattering of burning metal particles. Rigorous testing including discharge, hydraulic pressure, and burst pressure tests ensures safety, reliability, and effective performance, while anti-corrosive treatments and material conformity are strictly enforced.

**Applicability and Use:** Dry powder fire extinguishers for Class D fires are essential in industries and facilities where reactive metals such as magnesium, sodium, and potassium are handled or processed. They provide safe, targeted fire suppression for metal fires and should be operated only by trained personnel for maximum effectiveness and safety.

## IS 14951 : 2001 FIRE EXTINGUISHER — 135 LITRES CAPACITY MECHANICAL FOAM TYPE — SPECIFICATION

This standard specifies the requirements for a 135-litre mechanical foam fire extinguisher, focusing on materials, construction, charge, anti-corrosion treatment, and rigorous testing. The extinguisher body is made of mild steel, with components like the cap, hose, and fittings constructed from corrosion-resistant materials such as leaded tin bronze, brass, aluminium alloy, HDPE, and rubber, all conforming to relevant IS standards. The extinguishing medium is a foam concentrate mixed with water, propelled by a CO<sub>2</sub> gas cylinder. The design features a welded cylindrical body mounted on a wheeled carriage, complete with safety valves, level indicators, and efficient discharge fittings. Each extinguisher must discharge at least 90% of its capacity within 180 seconds, achieve a minimum throw of 12 metres, and pass stringent hydrostatic, performance, and corrosion resistance tests to ensure reliability and safety.

**Applicability and Use:** The 135-litre mechanical foam fire extinguisher is ideal for industrial sites, warehouses, and large storage areas where rapid and large-scale suppression of Class A and B fires especially those involving flammable liquids is critical. Its high capacity, extended discharge range, and mobile design make it suitable for emergency response in high-risk environments.

## IS 4862 (Part 1): 1986 SPECIFICATION FOR PORTABLE FIRE EXTINGUISHER FOR AIRCRAFT PART 1 HALON 1211 TYPE (FIRST REVISION) (Reaffirmed in December 2020)

This standard outlines the requirements for portable Halon 1211 fire extinguishers specifically designed for use by aircraft crew, detailing materials, construction, anti-corrosive treatment, and performance testing. The extinguisher body is made from high-quality steel, with critical components such as valves and nozzles crafted from leaded tin bronze or brass, and fittings from stainless steel, rubber, and copper, ensuring durability and corrosion resistance. It features an upright, squeeze-grip valve design, welded cylindrical body, pressure gauge, and mounting bracket, all constructed to withstand vibration and operational stresses in aircraft environments. The extinguisher undergoes rigorous testing including hydrostatic, leakage, drop, vibration, and burst pressure tests and must demonstrate reliable discharge performance across extreme temperatures. Uniform anti-corrosive coatings are required, and clear markings provide operational guidance and safety information.

**Applicability and Use:** These Halon 1211 extinguishers are intended for use by aircraft crew to combat Class B and C fires in confined, vibration-prone environments. Their compact design and clean agent discharge make them ideal for quick, safe fire suppression in aircraft cabins and cockpits.

## IS 2190: 2024 SELECTION, INSTALLATION AND MAINTENANCE OF FIRST-AID FIRE EXTINGUISHERS — PORTABLE AND MOBILE — CODE OF PRACTICE (*FIFTH REVISION*)

IS 2190:2024 provides comprehensive guidelines for the selection, installation, maintenance, and testing of portable and mobile (wheeled) first-aid fire extinguishers. The standard emphasizes that fire extinguishers serve as a first line of defence against small, incipient fires and are essential even in properties with fixed fire protection systems. It covers all major extinguisher types, ensuring they are appropriately chosen for specific fire hazards (Classes A, B, C, D, and F), properly installed for immediate accessibility, and regularly maintained by competent personnel. The standard also defines hazard classifications, minimum requirements for extinguisher placement and quantity, and stresses the importance of using ISI-marked extinguishers with consistent operation methods within a facility. Routine inspection, maintenance, and recharging procedures are detailed to ensure extinguishers remain reliable and effective throughout their service life.

**Applicability and Use:** Portable and mobile fire extinguishers, as guided by IS 2190, are intended for use in all types of buildings residential, commercial, industrial, and institutional to provide immediate response to small fires. Their proper selection, placement, and maintenance ensure rapid, effective fire control, protecting lives and property until professional firefighting services arrive.